Create the database BUS TRANSPORT and create given tables with all necessary constraints such as primary key, foreign key, unique key, not null and check constraints.

create database bus_transport;

create table train_master (train_no varchar(6),train_name varchar(25) not null,arrival_time date not null,depart_time date not null,no_hr decimal(5,2) not null,src_stn varchar(25) not null,end_stn varchar(25) not null,primary key(train_no),check(train_no like "%up" or train_no like "%dn"));

create table p_details (tick_no int(5),train_no varchar(6),seat_no int(2) not null,p_name varchar(35) not null,age int(2) not null,gender char(1) check (gender in ('M','F')),travel_dt date,class varchar(4) check (class in ('IA','IIA','IIIA','IC','II')),foreign key(train_no) references train_master(train_no) on delete cascade);

create table t_seat_master (train_no varchar(6),class varchar(4) check (class in ('IA','IIA','IIIA','IC','II')), total_s varchar(4) check (total_s>=25 AND total_s<=90), foreign key(train_no) references train_master(train_no) on delete cascade);

create table t_day_master (train_no varchar(6),day varchar(3) check (day in('MON','TUE','WED','THU','FRI','SAT','SUN')),foreign key(train_no) references train_master(train_no) on delete cascade);

Insert into train_master values(“3539UP”,”OKHAMAIL”,120000,220000,11.00,”Bombay”,”ahmedabad”);

Insert into train_mastervalues(“6354DN”,”GAREEBRATH”,120000,220000,35.00,”Baroda”,”banglore”),(“9848UP”,”JANTA XP”,080000,130000,03.00,”Rajkot”,”ahmedabad”);

insert t_day_master values("6354DN","SUN");

insert t_day_master values("9848UP","SUN");

insert p_details values(78955,”6354DN”,15,”RAJ PRATAP SINGH”,14,”M”,”2019-12-06”,”IA”);

insert into t_seat_master values("3539up","IIA",30),(“6354DN”,”IIA”,45),(“6354DN”,”IA”,50), (“6354DN”,”IC”,90), (“9848UP”,”IIA”,26);

Give all the train nanes starting from “Bombay” and going to “Ahmedabad” on Tuesday and Wednesday.

select t.train_no,t.src_stn,t.end_stn,d.day from train_master t,t_day_master d where (t.train_no=d.train_no and t.src_stn="bombay" and t.end_stn="ahmedabad")and(d.day="TUE" or d.day="WED");

```
mysql> select t.train_no,t.src_stn,t.end_stn,d.day from train_master t,t_day_master d where (t.train_no=d.train_no and t
.src_stn="bombay" and t.end_stn="ahmedabad")and(d.day="TUE" or d.day="WED");
+----------+---------+-----------+-----+
| train_no | src_stn | end_stn   | day |
+----------+---------+-----------+-----+
| 3539up   | bombay  | ahmedabad | WED |
+----------+---------+-----------+-----+
1 row in set (0.01 sec)
```

List all trains which is available on Sunday.
select t.train_no,t.train_name,d.day from train_master t,t_day_master d where (t.train_no=d.train_no and d.day="SUN");

```
mysql> select t.train_no,t.train_name,d.day from train_master t,t_day_master d where
 (t.train_no=d.train_no and d.day="SUN");
+----------+-------------+-----+
| train_no | train_name  | day |
+----------+-------------+-----+
| 6354DN   | GAREEB RATH | SUN |
| 9848UP   | JANTA EXP   | SUN |
+----------+-------------+-----+
2 rows in set (0.00 sec)
```

Give classwise seat availability on 10-June-2018 for train 9012DN.

List total seats classwise for train running on thrusday.
select t.train_no,t.train_name,c.class,c.total_s from train_master t,t_seat_master c,t_day_master d where (t.train_no=c.train_no and d.day="THU");

```
mysql> select t.train_no,t.train_name,c.class,c.total_s from train_master t,t_seat_m
aster c,t_day_master d where (t.train_no=c.train_no and d.day="THU");
+----------+-------------+-------+---------+
| train_no | train_name  | class | total_s |
+----------+-------------+-------+---------+
| 6354DN   | GAREEB RATH | IIA   | 45      |
| 6354DN   | GAREEB RATH | IA    | 50      |
| 6354DN   | GAREEB RATH | IC    | 90      |
+----------+-------------+-------+---------+
3 rows in set (0.00 sec)
```

List train names which have no sleeper class.
select train_no,class from t_seat_master where class<>"IC";

```
mysql> select train_no,class from t_seat_master where class<>"IC";
+----------+-------+
| train_no | class |
+----------+-------+
| 6354DN   | IIA   |
| 6354DN   | IA    |
| 3539up   | IIA   |
+----------+-------+
3 rows in set (0.00 sec)
```

List train number which run on Monday during 8:00: am to 1:00pm.
select t.train_no,t.src_stn,t.end_stn,d.day,t.arrival_time from train_master t,t_day_master d where (t.train_no=d.train_no and d.day="SUN") and (arrival_time between 080000 and 130000);

```
mysql> select t.train_no,t.src_stn,t.end_stn,d.day,t.arrival_time from train_master
t,t_day_master d where (t.train_no=d.train_no and  d.day="SUN") and (arrival_time be
tween 080000 and 130000);
+----------+---------+-----------+------+--------------+
| train_no | src_stn | end_stn   | day  | arrival_time |
+----------+---------+-----------+------+--------------+
| 9848UP   | Rajkot  | ahmedabad | SUN  | 08:00:00     |
+----------+---------+-----------+------+--------------+
1 row in set (0.00 sec)
```

Write a procedure which will print all train details going from Baroda to Banglore.

USE `bus_transport`;

DROP procedure IF EXISTS `bus_transport`.`btob`;

DELIMITER $$

USE `bus_transport`$$

CREATE DEFINER=`root`@`localhost` PROCEDURE `b2b`()

BEGIN

select * from train_master where src_stn="baroda" and end_stn="banglore";

END$$

DELIMITER ;

```
mysql> call b2b();
+---------+------------+--------------+-------------+-------+---------+----------+
| train_no | train_name  | arrival_time | depart_time | no_hr | src_stn | end_stn  |
+---------+------------+--------------+-------------+-------+---------+----------+
| 6354DN   | GAREEB RATH | 12:00:00     | 22:00:00    | 35.00 | Baroda  | banglore |
+---------+------------+--------------+-------------+-------+---------+----------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.04 sec)
```

SET 12

CUSTOMER(cid, fname, lname, city, country, phone)

create table customer(cid int not null,fname varchar(30) not null,lname varchar(30) not null,city varchar(30) not null,country varchar(30) not null,phone int(10) not null,primary key(cid));

ORDER (oid, oDate, oNumber, cid, oTotalAmount)

create table orders(oid int not null,odate date,onumber int not null,cid int ,ototal int not null,primary key(oid),foreign key(cid) references customer(cid));

1. List the number of customers in each country. Only include countries with more than 100 customers.
SELECT COUNT(cid), Country FROM customer GROUP BY Country HAVING COUNT(cid) > 2;

```
mysql> SELECT COUNT(cid), Country
    -> FROM customer
    -> GROUP BY Country
    -> HAVING COUNT(cid) > 2;
+------------+---------+
| COUNT(cid) | Country |
+------------+---------+
|          3 | florida |
+------------+---------+
1 row in set (0.00 sec)
```

2. List the number of customers in each country, except China, sorted high to low. Only include countries with 5 or more customers.
SELECT COUNT(cid),Country FROM customer where country<>"china" GROUP BY country HAVING COUNT(cid) > 2 order by count(cid) desc;

```
mysql> SELECT COUNT(cid),Country
    -> FROM customer
    -> where country<>"china"
    -> GROUP BY country
    -> HAVING COUNT(cid) > 2
    -> order by count(cid) desc;
+------------+---------+
| COUNT(cid) | Country |
+------------+---------+
|          3 | florida |
|          3 | india   |
+------------+---------+
2 rows in set (0.00 sec)
```

3. List all customers with average orders between Rs.5000 and Rs.6500.
SELECT customer.cid,customer.fname,avg(orders.ototal) as average_orders FROM customer left join orders on orders.cid = customer.cid WHERE EXISTS (SELECT ototal FROM orders WHERE orders.cid = customer.cid and ototal BETWEEN 5000 AND 6500) group by cid;

```
mysql> SELECT customer.cid,customer.fname,avg(orders.ototal) as average_orders
    -> FROM customer left join orders on orders.cid = customer.cid
    -> WHERE EXISTS
    -> (SELECT ototal FROM orders WHERE orders.cid = customer.cid and ototal BETWEEN
 5000 AND 6500)
    -> group by cid;
+-----+--------+----------------+
| cid | fname  | average_orders |
+-----+--------+----------------+
|   3 | tanish |      5700.0000 |
|   7 | rafik  |      5500.0000 |
+-----+--------+----------------+
2 rows in set (0.00 sec)
```

6. Create a procedure to display month names of dates of ORDER table. The month names should be unique.

USE `db`;

DROP procedure IF EXISTS `ShowMonth`;

DELIMITER $$

USE `db`$$

CREATE DEFINER=`root`@`localhost` PROCEDURE `ShowMonth`()

BEGIN

SELECT oid,monthname(odate) FROM orders group by monthname(odate);

END$$

DELIMITER ;

CALL ShowMonth();

```
mysql> call ShowMonth();
+-----+-----------------+
| oid | monthname(odate) |
+-----+-----------------+
|   1 | July            |
|   3 | May             |
|   4 | September       |
|   5 | February        |
|   6 | November        |
|   7 | December        |
+-----+-----------------+
6 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

SET 13

DISTRIBUTOR (dno, dname, daddress, dphone)

ITEM (itemno, itemname, colour, weight)

DIST_ITEM (dno, itemno, qty)

Create_DISTRIBUTOR Table:-

Query:

create table dis(dno int(5) primary key,dname varchar(30) not null,daddress varchar(50) not null,dphone int(30));

Output:

```
mysql> create table dis(dno int(5) primary key,dname varchar(30) not null,daddress varchar(50) not null,dphone int(30));
Query OK, 0 rows affected, 2 warnings (1.39 sec)
```

Create_ITEM Table:-

Query:

create table item(itemno int(5) primary key,itemname varchar(30),colour varchar(15),weight int(5));

Output:

```
mysql> create table item(itemno int(5) primary key,itemname varchar(30),colour varchar(15),weight int(5));
Query OK, 0 rows affected, 2 warnings (0.92 sec)
```

Create DIST_ITEM Table:-

Query:

create table dist_item(dno int(5) references dis,itemno int(5) references item,qty int(4),primary key(dno,itemno));

Output:

```
mysql> create table dist_item(dno int(5) references dis,itemno int(5) references item,qty int(4),primary key(dno,itemno));
Query OK, 0 rows affected, 3 warnings (0.59 sec)
```

Insert values in DISTRIBUTORS Table:-

Query:

insert into dis values

-> (10001,'Isha','Ahemdabad',1234567890),

-> (10002,'Bhargav','Mumbai',1876543210),

-> (10003,'Miloni','Pune',1692581470),

-> (10004,'vishal','Bhavnagar',1418529630),

-> (10005,'sakshi','Rajkot',1873216540);

Output:

```
mysql> insert into dis values
    ->   (10001,'Isha','Ahemdabad',1234567890),
    ->      (10002,'Bhargav','Mumbai',1876543210),
    ->      (10003,'Miloni','Pune',1692581470),
    ->      (10004,'vishal','Bhavnagar',1418529630),
    ->      (10005,'sakshi','Rajkot',1873216540);
Query OK, 5 rows affected (0.26 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

Insert values in ITEM Table:-

Query:

insert into item values

-> (101,'Table','Black',75),

-> (102,'Sofa','Red',90),

-> (103,'Deck chair','Gray',45),

-> (104,'Wing chair','Brown',35),

-> (105,'Park Bench','Green',45);

Output:

```
mysql> insert into item values
    -> (101,'Table','Black',75),
    -> (102,'Sofa','Red',90),
    -> (103,'Deck chair','Gray',45),
    -> (104,'Wing chair','Brown',35),
    -> (105,'Park Bench','Green',45);
Query OK, 5 rows affected (0.37 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

Insert values in DIST_ITEM Table:-

Query:

insert into dist_item values

-> (10001,101,2),

-> (10001,105,6),

-> (10002,102,4),

-> (10002,104,8),

-> (10003,105,7),

-> (10003,103,14),

-> (10004,101,4),

-> (10004,103,6),

-> (10005,104,5),

-> (10005,102,25);

Output:

```
mysql>  insert into dist_item values
    ->      (10001,101,2),
    ->      (10001,105,6),
    ->      (10002,102,4),
    -> (10002,104,8),
    -> (10003,105,7),
    -> (10003,103,14),
    -> (10004,101,4),
    -> (10004,103,6),
    -> (10005,104,5),
    -> (10005,102,25);
Query OK, 10 rows affected (0.16 sec)
```

Add a column CONTACT_PERSON to the DISTRIBUTOR table with the not null constraint.

Query:

alter table dis add(CONTACT_PERSON int(10) not null);

Output:

```
mysql> alter table dis add(CONTACT_PERSON int(10) not null);
Query OK, 0 rows affected, 1 warning (1.02 sec)
Records: 0  Duplicates: 0  Warnings: 1
```

Query:

select * from dis;

Output:

```
mysql> select * from dis;
+-------+---------+-----------+------------+----------------+
| dno   | dname   | daddress  | dphone     | CONTACT_PERSON |
+-------+---------+-----------+------------+----------------+
| 10001 | Isha    | Ahemdabad | 1234567890 |              0 |
| 10002 | Bhargav | Mumbai    | 1876543210 |              0 |
| 10003 | Miloni  | Pune      | 1692581470 |              0 |
| 10004 | vishal  | Bhavnagar | 1418529630 |              0 |
| 10005 | sakshi  | Rajkot    | 1873216540 |              0 |
+-------+---------+-----------+------------+----------------+
5 rows in set (0.12 sec)
```

Create a view LONDON_DIST on DIST_ITEM which contains only those records where distributors are from London. Make sure that this condition is checked for every DML against this

<u>Query:</u>

create view loan_dist as select d.dno,d.itemno,d.qty from dis di,dist_item d where di.daddress='Ahemdabad' and d.dno=di.dno;

<u>Output:</u>

```
mysql> create view loan_dist as select d.dno,d.itemno,d.qty from dis di,dist_item d where di.daddress='Ahemdabad' and d.dno=di.dno;
Query OK, 0 rows affected (0.31 sec)
```

<u>Query:</u>

select * from loan_dist;

<u>Output:</u>

```
mysql> select * from loan_dist;
+-------+--------+------+
| dno   | itemno | qty  |
+-------+--------+------+
| 10001 |    101 |    2 |
| 10001 |    105 |    6 |
+-------+--------+------+
2 rows in set (0.10 sec)
```

3)Display the details of all those items that have never been supplied.

<u>Query:</u>

select * from item where itemno not in(select itemno from dist_item);

<u>Output:</u>

```
mysql> select * from item where itemno not in(select itemno from dist_item);
Empty set (0.29 sec)
```

4)Delete all those items that have been supplied only once.

Query:

select * from dist_item;

Output:

```
mysql> select * from item where itemno not in(select itemno from dist_item);
Empty set (0.29 sec)
```

Query:

delete from dist_item where itemno in(select itemno from dist_item group by itemno having count(itemno)=1);

5) List the names of distributors who have an 'A' and also a 'B' somewhere in their names. Query:

select * from dis where dname like '%B%a%' or '%a%B%';

Output:

```
mysql> select * from dis where dname like '%B%a%' or '%a%B%';
+-------+---------+----------+------------+----------------+
| dno   | dname   | daddress | dphone     | CONTACT_PERSON |
+-------+---------+----------+------------+----------------+
| 10002 | Bhargav | Mumbai   | 1876543210 |              0 |
+-------+---------+----------+------------+----------------+
1 row in set, 1 warning (0.00 sec)
```

6) Count the number of items having the same colour but not having weight between 20 and 100.

Query:

select colour ,count(colour) from item where weight not between 20 and 100 group by colour;

Output:

```
mysql> select colour ,count(colour) from item where weight not between 20 and 100 group by colour;
+--------+---------------+
| colour | count(colour) |
+--------+---------------+
| Black  |             1 |
+--------+---------------+
1 row in set (0.12 sec)
```

7) Display all those distributors who have supplied more than 1000 parts of the same type.

Query:

select dname from dis where dno in(select dno from dist_item where qty>10);

Output:

```
mysql> select dname from dis where dno in(select dno from dist_item where qty>10);
+--------+
| dname  |
+--------+
| Miloni |
+--------+
1 row in set (0.00 sec)
```

8) Display the average weight of items of the same colour provided at least three items have That colour.

Query:

select colour,avg(weight) "Average Weight",count(colour) "Total no" from item group by colour having count(colour)>=2;

Output:

```
mysql> select colour,avg(weight) "Average Weight",count(colour) "Total no" from item group by colour having count(colour)>=2;
+--------+----------------+----------+
| colour | Average Weight | Total no |
+--------+----------------+----------+
| Black  |        95.0000 |        2 |
+--------+----------------+----------+
1 row in set (0.00 sec)
```

9)Display the position where a distributor name has an 'OH' in its spelling somewhere after the fourth character

Query:

select dname,instr(dname,"sh") as POSITION from dis where instr(dname,'sh')<>0;

Output:

```
mysql>  select dname,instr(dname,"sh") as POSITION from dis where instr(dname,'sh')<>0;
+--------+----------+
| dname  | POSITION |
+--------+----------+
| Isha   |        2 |
| vishal |        3 |
| sakshi |        4 |
+--------+----------+
3 rows in set (0.01 sec)
```

10) Count the number of distributors who have a phone connection and are supplying item number 'I100'.

Query:

select * from dis d,dist_item di where d.dno=di.dno and di.itemno=105 and d.dphone is not null;

Output:

```
mysql> select * from dis d,dist_item di where d.dno=di.dno and di.itemno=105 and d.dphone is not null;
+-------+--------+-----------+------------+----------------+-------+--------+------+
| dno   | dname  | daddress  | dphone     | CONTACT_PERSON | dno   | itemno | qty  |
+-------+--------+-----------+------------+----------------+-------+--------+------+
| 10001 | Isha   | Ahemdabad | 1234567890 |              0 | 10001 |    105 |    6 |
| 10003 | Miloni | Pune      | 1692581470 |              0 | 10003 |    105 |    7 |
+-------+--------+-----------+------------+----------------+-------+--------+------+
2 rows in set (0.62 sec)
```

11) Create a view on the tables in such a way that the view contains the distributor name, item name and the quantity supplied.

Query:

create view dis_view as select DNAME,ITEMNAME,QTY from dis d,item i,dist_item di where d.dno=di.dno and di.itemno=i.itemno;

Output:

```
mysql> select * from dis d,dist_item di where d.dno=di.dno and di.itemno=105 and d.dphone is not null;
+-------+--------+-----------+------------+----------------+-------+--------+------+
| dno   | dname  | daddress  | dphone     | CONTACT_PERSON | dno   | itemno | qty  |
+-------+--------+-----------+------------+----------------+-------+--------+------+
| 10001 | Isha   | Ahemdabad | 1234567890 |              0 | 10001 |    105 |    6 |
| 10003 | Miloni | Pune      | 1692581470 |              0 | 10003 |    105 |    7 |
+-------+--------+-----------+------------+----------------+-------+--------+------+
2 rows in set (0.62 sec)
```

Query:

select * from dis_view;

Output:

```
mysql> select * from dis_view;
+---------+------------+------+
| DNAME   | ITEMNAME   | QTY  |
+---------+------------+------+
| Isha    | Table      |    2 |
| Isha    | Banch      |    6 |
| Bhargav | Sofa       |    4 |
| Bhargav | Wing chair |    8 |
| Miloni  | Deck chair |   14 |
| Miloni  | Banch      |    7 |
| vishal  | Table      |    4 |
| vishal  | Deck chair |    6 |
| sakshi  | Wing chair |    5 |
+---------+------------+------+
9 rows in set (0.10 sec)
```

12) List the name, address and phone number of distributors who have the same three digits in their number as 'Mr. Talkative'.

Query:

select dname,dphone,daddress from dis where substr(dphone,1,3) in (select substr(dphone,1,3) from dis where dname="Isha") and dname!="Isha";

Output:

```
mysql> select dname,dphone,daddress from dis where substr(dphone,1,3) in (select substr(dphone,1,3) from dis where dname="Isha") and dname!="Isha";
Empty set (2.41 sec)
```


13) List all distributor names who supply either item I1 or I7 or the quantity supplied is more than 100.

Query:

select DNAME from dist_item i, dis d where d.dno = i.dno and i.itemno in (102,103) and qty>=10 group by dname;

Output:

```
mysql>  select DNAME from dist_item i, dis d where d.dno = i.dno and i.itemno in (102,103)  and qty>=10 group by dname;
+--------+
| DNAME  |
+--------+
| Miloni |
+--------+
1 row in set (0.05 sec)
```

14) Display the data of the top three heaviest ITEMS

Query:

select * from item order by weight desc limit 3;

Output:

```
mysql> select * from item order by weight desc limit 3;
+--------+----------+--------+--------+
| itemno | itemname | colour | weight |
+--------+----------+--------+--------+
|    105 | Banch    | Black  |    115 |
|    102 | Sofa     | Red    |     90 |
|    101 | Table    | Black  |     75 |
+--------+----------+--------+--------+
3 rows in set (0.03 sec)
```

Set-14

d) Write a procedure to display top five highest paid workers who are specialized in "PAINTING" using table WORKER (workerid, name, wage_per_hour, specialized_in, manager_id)

<u>Query to create table:</u>

create table worker(wid int(3) primary key,name varchar(30),wages_per_hour int(5),spe_in varchar(30),man_id int(3));

<u>Output:</u>

```
mysql> create table worker(wid int(3) primary key,name varchar(30),wages_per_hour int(5),spe_in varchar(30),man_id int(3
));
Query OK, 0 rows affected, 3 warnings (1.19 sec)
```

<u>Query to insert data into table:</u>

insert into worker values(101,'Isha',500,'painting',1001),(102,'Yash',200,'Roof',1001),(103,'Vivek',200,'painting',1001),(104,'Ria',230,'painting',1002),(105,'Diya',560,'carpantry',1003),(106,'Ramu',450,'painting',1004);

<u>Output:</u>

```
mysql> insert into worker values(101,'Isha',500,'painting',1001),(102,'Yash',200,'Roof',1001),(103,'Vivek',200,'painting
',1001),(104,'Ria',230,'painting',1002),(105,'Diya',560,'carpantry',1003),(106,'Ramu',450,'painting',1004);
Query OK, 6 rows affected (0.29 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

<u>procedure :</u>

CREATE DEFINER=`root`@`localhost` PROCEDURE `worker`()

BEGIN

select name from worker where spe_in='painting' order by wages_per_hour desc limit 5;

END

Output:

```
CREATE DEFINER=`root`@`localhost` PROCEDURE `worker`()
BEGIN

    select name from worker where spe_in='painting' order by wages_per_hour de


END
```

```
mysql> call worker();
+-------+
| name  |
+-------+
| Isha  |
| Ramu  |
| Ria   |
| Vivek |
+-------+
4 rows in set (0.01 sec)
```

b) Write a function that returns total number of incomplete jobs, using table JOB (jobid, type_of_job, status)

Query to create table:

create table job(jobid int(3) primary key,type_of_job varchar(30),status   varchar(30));

function :

```
CREATE DEFINER=`root`@`localhost` FUNCTION `job`(st varchar(10)) RETURNS
varchar(11) CHARSET utf8mb4

    DETERMINISTIC

BEGIN

        declare sst varchar(10);

     select count(status) into sst from jo where status='incomplete';

RETURN sst ;

END
```

Output:

```
CREATE DEFINER=`root`@`localhost` FUNCTION `job`(st varchar(10)) RETURNS varch
    DETERMINISTIC
BEGIN
    declare sst varchar(10);
     select count(status) into sst from jo where status='incomplete';
RETURN sst ;
END
```

```
mysql> select job(status) from jo;
+-------------+
| job(status) |
+-------------+
| 2           |
| 2           |
| 2           |
+-------------+
3 rows in set (0.00 sec)
```

c) Write a function which displays the number of items whose weight fall between a given ranges for a particular color using table ITEM (itemno, name, color, weight)

<u>Query to create table:</u>

```
create table item(itemno int(3) primary key,name  varchar(30),color
varchar(30),weight int(5));
```

```
CREATE DEFINER=`root`@`localhost` FUNCTION `im`(we integer(5)) RETURNS int(11)

   DETERMINISTIC

BEGIN

        declare ino integer(3);

   select weight into we from item where   weight>=10 and weight<=25 and
itemno=ino   ;

RETURN we;


END
```

```
CREATE DEFINER=`root`@`localhost` FUNCTION `im`(we integer(5)) RETURNS int(11)
    DETERMINISTIC
BEGIN
    declare ino integer(3);
    select weight into we from item where   weight>=10 and weight<=25 and item
RETURN we;

END
```

```
mysql> select name,im(weight) from item where color='Black';
+--------+------------+
| name   | im(weight) |
+--------+------------+
| TV     |         25 |
| laptop |         15 |
| chair  |         25 |
+--------+------------+
3 rows in set (0.00 sec)
```

SET-15

EMP (empno, empnm, empadd, salary, date_birth, joindt, deptno)
DEPT (deptno, deptnm)
Write a PL/SQL block (table above EMP-DEPT table) which takes as input Department name and
displays all the employees of this department who has been working since last five years

1.Create Database SET15;

Query:-create database SET15;

Output:-

```
mysql> create database SET15;
Query OK, 1 row affected (0.15 sec)
```

use SET15;

```
mysql> use SET15;
Database changed
```

2.Create Table DEPT:

Query:-create table DEPT(deptno int(5) PRIMARY KEY,deptnm varchar(20) NOT NULL);

Output:-

```
mysql> create table DEPT(deptno int(5) PRIMARY KEY,deptnm varchar(20) NOT NULL);
Query OK, 0 rows affected, 1 warning (0.54 sec)
```

3.Create Table EMP:

Query: -CREATE table EMP(empno int(5) PRIMARY KEY,empnm varchar(20),empadd varchar(50),salary int,date_birth date,joindt date,deptno int(5),FOREIGN KEY(deptno) REFERENCES DEPT(deptno));

Output: -

```
mysql> CREATE table EMP(empno int(5) PRIMARY KEY,empnm varchar(20),empadd varchar(50),salary int,dat
e_birth date,joindt date,deptno int(5),FOREIGN KEY(deptno) REFERENCES DEPT(deptno));
Query OK, 0 rows affected, 2 warnings (1.34 sec)
```

4.Describe both databases;

Query: -DESC DEPT;

Output: -

```
mysql> DESC DEPT;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| deptno | int(5)      | NO   | PRI | NULL    |       |
| deptnm | varchar(20) | NO   |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
2 rows in set (0.01 sec)
```

Query: -DESC EMP;

Output: -

```
mysql> DESC EMP;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| empno      | int(5)      | NO   | PRI | NULL    |       |
| empnm      | varchar(20) | YES  |     | NULL    |       |
| empadd     | varchar(50) | YES  |     | NULL    |       |
| salary     | int(11)     | YES  |     | NULL    |       |
| date_birth | date        | YES  |     | NULL    |       |
| joindt     | date        | YES  |     | NULL    |       |
| deptno     | int(5)      | YES  | MUL | NULL    |       |
+------------+-------------+------+-----+---------+-------+
7 rows in set (0.01 sec)
```

5.INERT Data In Tables

Query: -INSERT INTO DEPT values("101","Computer"),("102","Mechanical");

Output: -

```
mysql> INSERT INTO DEPT values("101","Computer"),("102","Mechanical");
Query OK, 2 rows affected (0.17 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

Query: -INSERT INTO EMP values(1,"Bhargav","Ahemdabad","50000","1998/05/14","2012/12/01",101),(2,"Bhargav","Mumbai","60000","1996/12/05","2012/11/10",102),(3,"Vishal","Ahemdabad","45000","1998/02/11","2011/01/01",101);

Output: -

```
mysql> INSERT INTO EMP values(1,"Bhargav","Ahemdabad","50000","1998/05/14","2012
rgav","Mumbai","60000","1996/12/05","2012/11/10",102),(3,"Vishal","Ahemdabad","4
2011/01/01",101);
Query OK, 3 rows affected (0.17 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

5.Disply Data

Query: -SELECT * from DEPT;

Query: -SELECT * from EMP;

Output: -

```
mysql> SELECT * from DEPT;
+--------+------------+
| deptno | deptnm     |
+--------+------------+
|    101 | Computer   |
|    102 | Mechanical |
+--------+------------+
2 rows in set (0.00 sec)

mysql> SELECT * from EMP;
+-------+---------+-----------+--------+------------+------------+--------+
| empno | empnm   | empadd    | salary | date_birth | joindt     | deptno |
+-------+---------+-----------+--------+------------+------------+--------+
|     1 | Bhargav | Ahemdabad |  50000 | 1998-05-14 | 2012-12-01 |    101 |
|     2 | Bhargav | Mumbai    |  60000 | 1996-12-05 | 2012-11-10 |    102 |
|     3 | Vishal  | Ahemdabad |  45000 | 1998-02-11 | 2011-01-01 |    101 |
+-------+---------+-----------+--------+------------+------------+--------+
3 rows in set (0.00 sec)
```

6.Create Procedure

Query:

DELIMITER $$

CREATE PROCEDURE FindEmp(IN dname TEXT)

BEGIN

SELECT e.empnm,d.deptnm FROM dept d,emp e WHERE e.deptno=d.deptno AND d.deptnm=dname AND DATEDIFF(CURDATE(), e.joindt)/365 > 5;

END$$

Output:

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE FindEmp(IN dname TEXT)
    -> BEGIN
    -> SELECT e.empnm,d.deptnm FROM dept d,emp e WHERE e.deptno=d.deptno AND d.deptnm=dname AND
 DATEDIFF( CURDATE(), e.joindt )/365 > 5;
    -> END$$
Query OK, 0 rows affected (0.23 sec)
```

7.Call Procedure:

Query: - call FindEmp("Computer")

Output: -

```
mysql> call FindEmp("Computer")
    -> ;
+----------+-----------+
| empnm    | deptnm    |
+----------+-----------+
| Bhargav  | Computer  |
| Vishal   | Computer  |
+----------+-----------+
2 rows in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)
```

SET-16

EMPMAST (empno, name, pfno, empbasic, deptno, designation)
DEPT (DNO, DNAME)
Rules: HRA = 15% of basic
DA = 50% of basic
Medical = 100
PF = 8.33%of basic
Print Salary slip. Design your own format.

1.Create Database

Query: -CREATE DATABASE set16;

Output: -

```
mysql> create database set16;
Query OK, 1 row affected (0.18 sec)
```

Query: -USE set16;

Output: -

```
mysql> use set16
Database changed
```

2.Create Table EMPMAST

Query: - CREATE TABLE EMPMAST(empno int(5),name varchar(30),pfno int(5),empbasic float,deptno int(5) PRIMARY KEY,designation varchar(30))

Output: -

```
mysql> CREATE TABLE EMPMAST(empno int(5),name varchar(30),pfno int(5),empbasic f
loat,deptno int(5) PRIMARY KEY,designation varchar(30));
Query OK, 0 rows affected, 3 warnings (1.07 sec)
```

3.Create Table DEPT

Query: - CREATE TABLE DEPT(deptno int(5) REFERENCES EMPMAST(deptno),dname varchar(30))

Output: -

```
mysql> CREATE TABLE DEPT(deptno int(5) REFERENCES EMPMAST(deptno),dname varchar(
30))
    -> ;
Query OK, 0 rows affected, 1 warning (1.37 sec)
```

4.Describe Tables;

Query: - DESC EMPMAST;

Query: -DESC DEPT;

Output: -

```
mysql> DESC EMPMAST;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| empno       | int(5)      | YES  |     | NULL    |       |
| name        | varchar(30) | YES  |     | NULL    |       |
| pfno        | int(5)      | YES  |     | NULL    |       |
| empbasic    | float       | YES  |     | NULL    |       |
| deptno      | int(5)      | NO   | PRI | NULL    |       |
| designation | varchar(30) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
6 rows in set (0.08 sec)

mysql> DESC DEPT;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| deptno | int(5)      | YES  |     | NULL    |       |
| dname  | varchar(30) | YES  |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
2 rows in set (0.01 sec)
```

5.Insert Data Into EMPMAST

Query: - insert into EMPMAST values(101,"Bhargav",11111,50000,1,"Manager"),(102,"Hiren",22222,60000,2,"Manager");

Output: -

```
mysql> insert into EMPMAST values(101,"Bhargav",11111,50000,1,"Manager"),(102,"H
iren",22222,60000,2,"Manager");
Query OK, 2 rows affected (0.12 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

Query: - insert into DEPT values(1,"HR"),(2,"IT");

Output: -

```
mysql> insert into DEPT values(1,"HR"),(2,"IT");
Query OK, 2 rows affected (0.21 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

6.Show Table Data

Query: -select * from EMPMAST;

Query: -select * from DEPT;

Output: -

```
mysql> select * from EMPMAST;
+-------+---------+-------+----------+--------+-------------+
| empno | name    | pfno  | empbasic | deptno | designation |
+-------+---------+-------+----------+--------+-------------+
|   101 | Bhargav | 11111 |    50000 |      1 | Manager     |
|   102 | Hiren   | 22222 |    60000 |      2 | Manager     |
+-------+---------+-------+----------+--------+-------------+
2 rows in set (0.07 sec)

mysql> select* from DEPT;
+--------+-------+
| deptno | dname |
+--------+-------+
|      1 | HR    |
|      2 | IT    |
+--------+-------+
2 rows in set (0.07 sec)
```

7.Print Salary Slip

Query: - SELECT name,empbasic,(empbasic*15/100) as HRA,(empbasic*50/100) as DA,(100) as Medical,(empbasic*8.33/100) as PF,empbasic+(empbasic*15/100)+(empbasic*50/100)+(100)-(empbasic*8.33/100) as Final FROM EMPMAST;

Output: -

```
mysql> SELECT name,empbasic,(empbasic*15/100) as HRA,(empbasic*50/100) as DA,(10
0) as Medical,(empbasic*8.33/100) as PF,empbasic+(empbasic*15/100)+(empbasic*50/
100)+(100)-(empbasic*8.33/100) as Final FROM EMPMAST;
+---------+----------+------+-------+---------+------+-------+
| name    | empbasic | HRA  | DA    | Medical | PF   | Final |
+---------+----------+------+-------+---------+------+-------+
| Bhargav |    50000 | 7500 | 25000 |     100 | 4165 | 78435 |
| Hiren   |    60000 | 9000 | 30000 |     100 | 4998 | 94102 |
+---------+----------+------+-------+---------+------+-------+
2 rows in set (0.00 sec)
```

<u>**SET-17**</u>

QUESTION

Consider the Bank schema as

ACCOUNT (AC_NO, NAME, AC_TYPE, BALANCE_AMT, BALANCE_DATE)

TRANSACTION (AC_NO, DATE, TR_TYPE, AMOUNT, PREV_BALANCE,

REMARK)

Note: 1. AC_type may be S for saving or C for current, 2. TR_type may be D for deposit or W for withdrawal.

> Write a procedure to print the Bank Transaction details by passing from and to dates.

QUERY

DELIMITER $$

CREATE PROCEDURE p1(IN date1 date, IN date2 date)

BEGIN

SELECT * from transaction where date between date1 AND date2;

END$$

call P1("2019/01/01","2019/01/04")

OUTPUT

Account Table

```
mysql> select * from account;
+-------+---------+---------+-------------+--------------+
| ac_no | name    | ac_type | balance_amt | balance_date |
+-------+---------+---------+-------------+--------------+
|   101 | Vishal  | S       |       50000 | 2019-04-12   |
|   102 | Bhargav | C       |       30000 | 2019-05-11   |
|   103 | Guruji  | C       |       20000 | 2018-01-01   |
|   104 | Param   | S       |       40000 | 2017-03-02   |
+-------+---------+---------+-------------+--------------+
4 rows in set (0.49 sec)
```

Transaction Table

```
mysql> select * from transaction;
+-------+------------+---------+--------+--------------+--------+
| ac_no | date       | tr_type | amount | prev_balance | remark |
+-------+------------+---------+--------+--------------+--------+
|   102 | 2019-01-01 | D       |  20000 |        10000 |        |
|   101 | 2019-01-02 | D       |  10000 |        10000 |        |
|   104 | 2019-01-03 | W       |  30000 |        40000 |        |
|   103 | 2019-01-04 | W       |  10000 |        20000 |        |
+-------+------------+---------+--------+--------------+--------+
4 rows in set (0.07 sec)
```

```
mysql> call P1("2019/01/01","2019/01/04")
    -> ;
+-------+------------+---------+--------+--------------+--------+
| ac_no | date       | tr_type | amount | prev_balance | remark |
+-------+------------+---------+--------+--------------+--------+
|   102 | 2019-01-01 | D       |  20000 |        10000 |        |
|   101 | 2019-01-02 | D       |  10000 |        10000 |        |
|   104 | 2019-01-03 | W       |  30000 |        40000 |        |
|   103 | 2019-01-04 | W       |  10000 |        20000 |        |
+-------+------------+---------+--------+--------------+--------+
4 rows in set (0.25 sec)
```

<u>**SET 18**</u>

BRANCH (branch_no, area, city)

MEMBERS (mno, name branch_no, salary, manager_no)

create table Branch(branch_no int primary key,area varchar(20),city varchar(20));

```
mysql> select *from branch;
+-----------+--------+-------+
| branch_no | area   | city  |
+-----------+--------+-------+
|       101 | vasna  | satna |
|       102 | paldi  | patna |
|       103 | sola   | ratna |
|       104 | chola  | matna |
|       105 | khasna | watna |
|       106 | champa | batna |
+-----------+--------+-------+
6 rows in set (0.02 sec)
```

create table Members(mno int primary key,name varchar(15),branch_no int,salary int,manager_no int,foreign key(branch_no) references Branch(branch_no));

```
mysql> select *from members;
+-----+---------+-----------+--------+------------+
| mno | name    | branch_no | salary | manager_no |
+-----+---------+-----------+--------+------------+
|   1 | tanish  |       101 |   4434 |          7 |
|   2 | param   |       101 |  54453 |          3 |
|   3 | pavan   |       102 | 254543 |         11 |
|   4 | bhargav |       102 |  94543 |         10 |
|   5 | vishal  |       103 |  54533 |          9 |
|   6 | ankit   |       104 |   2500 |          2 |
|   7 | jeet    |       103 |  24545 |          3 |
|   8 | yagnik  |       105 |  24545 |          6 |
|   9 | darshan |       106 |   5454 |          4 |
|  10 | farukh  |       103 |  45432 |          5 |
|  11 | mr x    |       105 | 454545 |          1 |
+-----+---------+-----------+--------+------------+
11 rows in set (0.00 sec)
```

.

1. Write a procedure which list the name of members who earns more than that of his managers.

CREATE DEFINER=`root`@`localhost` PROCEDURE `a18`()

BEGIN

select m1.name,m1.salary,m.name,m.salary from members m,members m1 where m1.manager_no=m.mno and m1.salary>m.salary;

```
mysql> call a18();
+---------+--------+---------+--------+
| name    | salary | name    | salary |
+---------+--------+---------+--------+
| bhargav |  94543 | farukh  |  45432 |
| vishal  |  54533 | darshan |   5454 |
| yagnik  |  24545 | ankit   |   2500 |
| mr x    | 454545 | tanish  |   4434 |
+---------+--------+---------+--------+
4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

2. Write a procedure which gives details of employee having maximum salary branch wise.

CREATE DEFINER=`root`@`localhost` PROCEDURE `b18`()

BEGIN

select max(salary),branch_no,name from members group by branch_no;

END

```
mysql> call b18();
+-------------+-----------+---------+
| max(salary) | branch_no | name    |
+-------------+-----------+---------+
|       54453 |       101 | tanish  |
|      254543 |       102 | pavan   |
|       54533 |       103 | vishal  |
|        2500 |       104 | ankit   |
|      454545 |       105 | yagnik  |
|        5454 |       106 | darshan |
+-------------+-----------+---------+
6 rows in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)
```
